

Supplementary Material: Within- and between-year seasonal patterns of respiratory pathogens and the potential for concurrent outbreaks

Keywords: respiratory pathogens, influenza, RSV, seasonal dynamics, survival analysis, concurrent outbreaks, geographic variation

1. Supplemental Materials

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1.1. Tables

1.1.1. Comparable Models by City and Pathogen

Table 1: Table S1: Number of comparable models ($\Delta AIC < 7$) per city and pathogen. This table illustrates the distribution of model availability across the eight study cities, highlighting the variation in the number of comparable models per city.

City	Influenza	RSV
<i>North</i>		
Beijing	20	39
Lanzhou	2	4
Xian	2	3
<i>North Total</i>	24	46
<i>South</i>		
Wuhan	20	22
Suzhou	18	3
Guangzhou	4	11
Wenzhou	2	2
Yunfu	2	2
<i>South Total</i>	46	40
Grand Total	70	86

1.1.2. Median Probability of Concurrent Outbreaks by Risk Threshold

Table 2: Table S2: Median probability of concurrent outbreaks by city and risk threshold. Values represent the median proportion of years with concurrent high-risk periods for both Influenza and RSV across 1000 simulation paths. Risk thresholds are defined as joint top 30%, 20%, and 10% percentiles of disease burden.

City	Top 30%	Top 20%	Top 10%
<i>North</i>			
Beijing	0.200	0.100	0.000
Lanzhou	0.118	0.059	0.000
Xian	0.450	0.350	0.150
<i>South</i>			
Wuhan	0.400	0.300	0.150
Suzhou	0.350	0.200	0.100
Guangzhou	0.188	0.125	0.062
Wenzhou	0.812	0.500	0.250
Yunfu	0.375	0.188	0.125

1.1.3. Median Time to Next Concurrent Outbreak by Risk Threshold

Table 3: Table S3: Median time to subsequent concurrent outbreak (years) by city and risk threshold, with 95% confidence intervals. Values represent the x-intercepts of the dashed vertical lines in Figure 6, indicating the time (in years) at which 50% of simulation paths are expected to experience their subsequent concurrent outbreak. Confidence intervals (in parentheses) reflect uncertainty in the Kaplan-Meier survival estimates, which properly account for right-censoring of paths that never experience a second outbreak. The relatively narrow confidence intervals reflect high precision from aggregating across 1000 simulation paths per city. Values marked as “—” indicate that the 50% threshold was not reached within the simulation window. Risk thresholds are defined as joint top 30%, 20%, and 10% percentiles of disease burden.

City	Top 30%	Top 20%	Top 10%
<i>North</i>			
Beijing	3.0 (2.0–3.0)	5.0 (4.0–6.0)	—
Lanzhou	5.0 (5.0–6.0)	—	—
Xian	1.0 (1.0–1.0)	2.0 (2.0–2.0)	3.0 (3.0–3.0)
<i>South</i>			
Wuhan	1.0 (1.0–1.0)	2.0 (2.0–2.0)	3.0 (3.0–3.0)
Suzhou	2.0 (2.0–2.0)	3.0 (3.0–3.0)	6.0 (5.0–6.0)
Guangzhou	3.0 (3.0–4.0)	6.0 (6.0–7.0)	—
Wenzhou	1.0 (1.0–1.0)	1.0 (1.0–1.0)	2.0 (2.0–2.0)
Yunfu	2.0 (2.0–2.0)	3.0 (3.0–3.0)	7.0 (6.0–8.0)

1.1.4. Optimal Within-Year and Between-Year Cycle Lengths

Table 4: Table S4: Optimal within-year and between-year cycle lengths (months) for each city and pathogen combination. Values represent the cycle lengths of the best-performing model (lowest AIC) for each city-pathogen pair. Within-year cycles are shown for both pathogens; between-year cycles are only shown for models that included multi-seasonal decomposition (MSTL). These values correspond to the data presented in Figure 3 (A) and (B) of the main text.

Region	City	Within-year cycle (months)		Between-year cycle (months)	
		Influenza	RSV	Influenza	RSV
North					
	Beijing	13.0	10.0	28.0	24.0
	Lanzhou	13.0	13.0	20.0	35.0
	Xian	13.0	13.0	36.0	33.0
South					
	Suzhou	14.0	12.0	26.0	21.0
	Wuhan	13.0	12.0	30.0	19.0
	Wenzhou	12.0	12.0	22.0	18.0
	Guangzhou	13.0	12.0	34.0	31.0
	Yunfu	14.0	10.0	21.0	24.0

1.2. Figures

1.2.1. Distribution of Between-Year Cycle Lengths

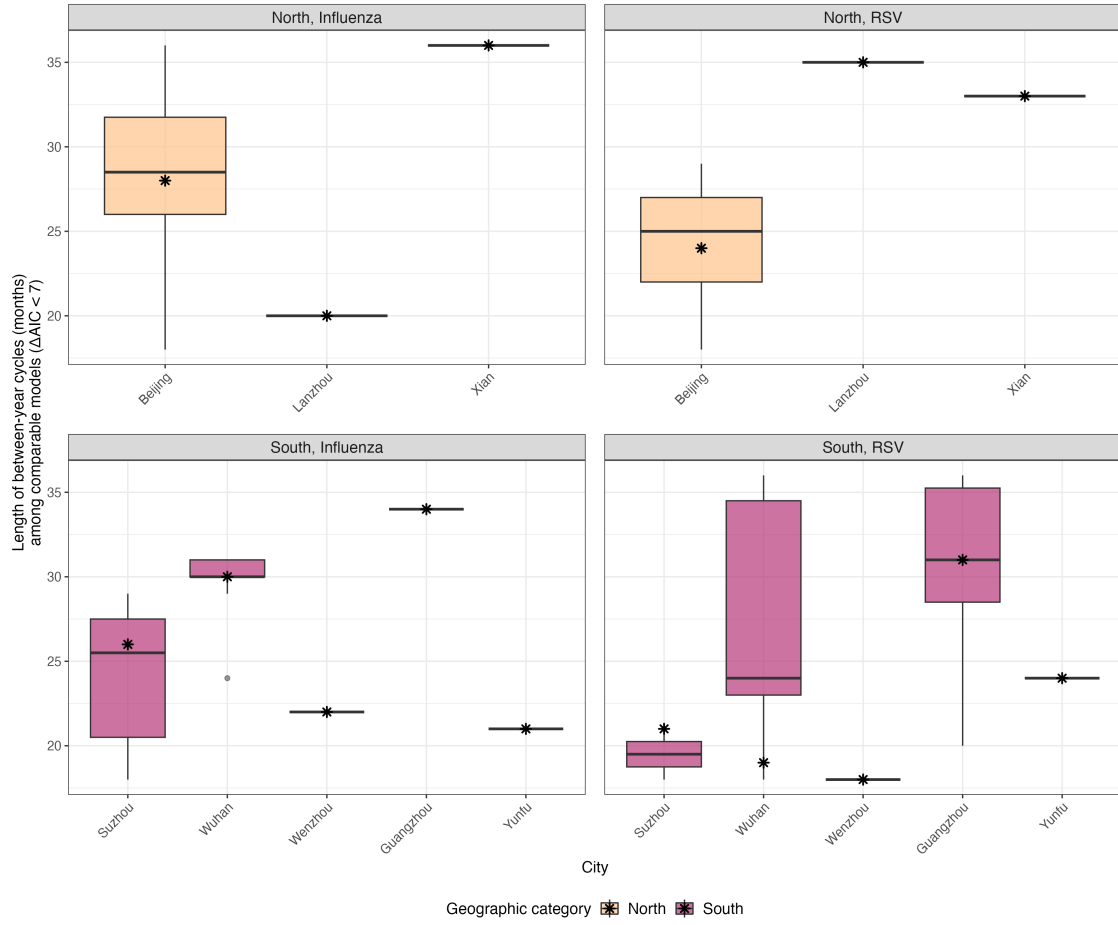


Figure 1: Figure S1: Distribution of between-year cycle lengths (months) among comparable models ($\Delta AIC < 7$) by city and pathogen, separated by geographic region (North vs South). Each panel displays boxplots of the between-year cycle lengths (in months) for each city within the specified region and pathogen combination. The asterisk (*) marks the best-performing model (lowest AIC) for each city-pathogen combination.

Supplementary Text 1. Literature search strategy

To support modelling of both within-year seasonality and between-year (interannual) variation, we implemented a unified data assembly strategy for both respiratory syncytial virus (RSV) and influenza virus (InfV). We required year-indexed, continuous multi-year time series (e.g., data points labelled 2017-01, 2017-02, ...), and strictly excluded studies that only reported aggregated seasonal profiles obtained by averaging across years within each calendar month or epidemiological week (i.e., month-of-year or week-of-year values without a year index).

RSV datasets were identified from an existing systematic review [2], while influenza datasets were assembled for the resulting RSV-anchored cities using a hybrid approach involving both systematic review screening and targeted literature searching.

RSV dataset identification

Candidate RSV studies were obtained from the extraction table released with the systematic review by Guo et al. [2] (RSVseasonality_China_2019_foruploading.xlsx; available at https://github.com/Glinnnng/RSVseasonality_China/blob/master/RSVseasonality_China_2019_foruploading.xlsx; 161 records). Records were first stratified by study setting, yielding 142 mainland China records after excluding Taiwan (10) and Hong Kong (9).

We parsed the study periods and filtered these records using the following eligibility criteria: prefecture-/city-level surveillance (administrative level two), study duration of at least five consecutive years (≥ 60 months), and monthly or weekly temporal resolution. Applying the duration filter (≥ 5 years) retained 19 mainland records. Further restricting these to city-level surveillance yielded 13 candidate records.

These 13 candidates were manually reviewed to verify the time-series structure. Consistent with our general strategy, four records reporting only multi-year month-of-year averages or insufficient temporal resolution (e.g., quarterly data) were excluded. This screening process resulted in nine eligible continuous time series covering eight unique cities.

While most cities contributed a single unique dataset, Beijing had two eligible time series. For Beijing, we selected the dataset covering the longer time span (7 years vs. 5 years) to maximize information on interannual variation. The final RSV dataset comprised eight cities: Beijing, Lanzhou, Xi'an, Wuhan, Suzhou, Wenzhou, Guangzhou, and Yunfu. The complete data assembly process for both RSV and influenza is summarized in Supplementary Figure S2.

Influenza dataset identification

Influenza time series were assembled for the same eight RSV-anchored cities. We prioritized datasets from existing systematic reviews before conducting targeted searches.

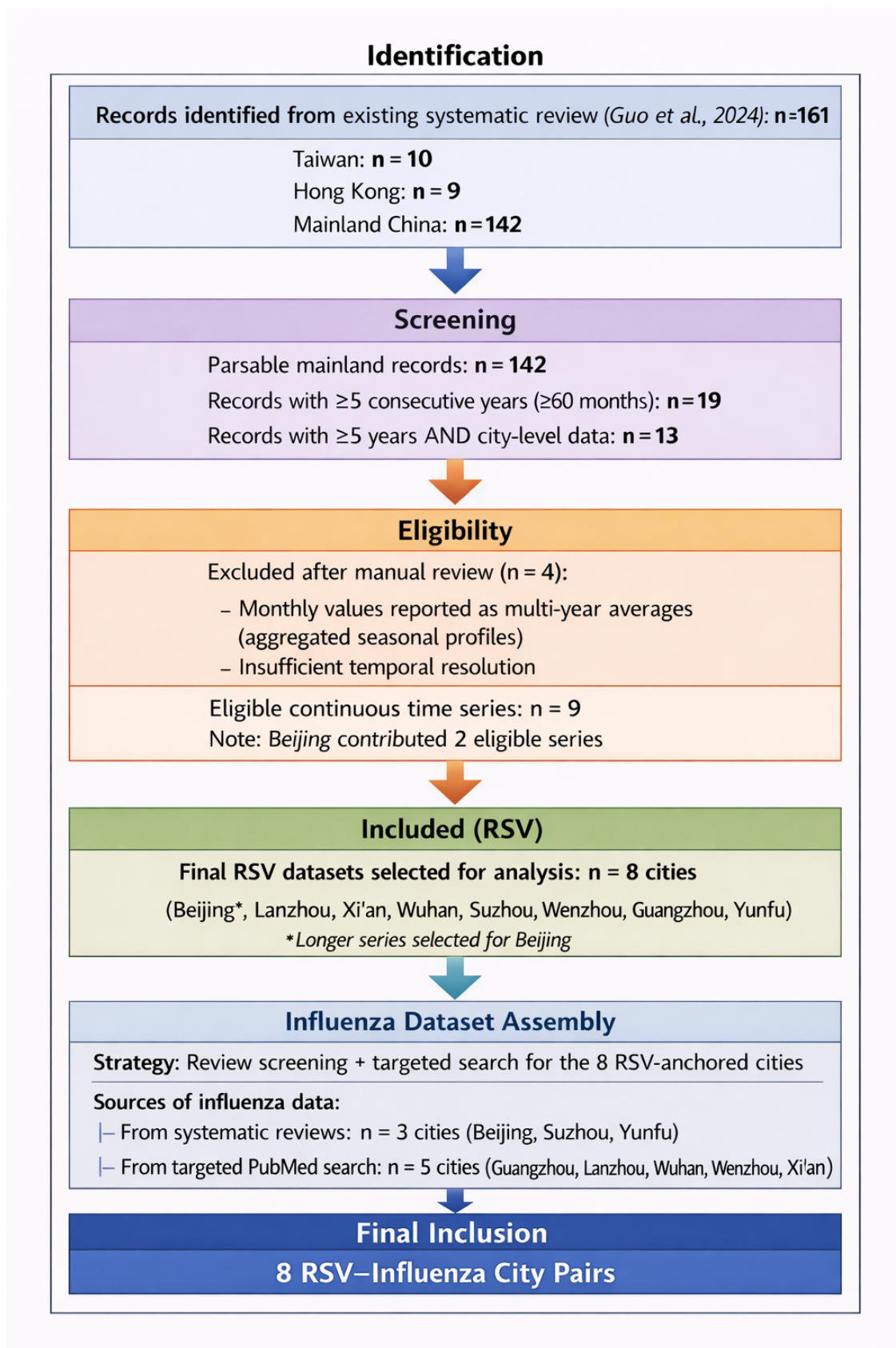


Figure 2: Supplementary Figure S2. Flowchart of data assembly and study selection. The diagram illustrates the stepwise screening process for RSV surveillance data (top) and the subsequent assembly of matched influenza time series for the identified cities (bottom).

First, we screened two major influenza systematic reviews [1, 3] for eligible city-level data. This process identified suitable time series for three cities:

- **Beijing** and **Suzhou**: Datasets were identified from Li et al. [3].
- **Yunfu**: Dataset was identified from Diamond et al. [1].

For the remaining five cities (Guangzhou, Lanzhou, Wuhan, Wenzhou, and Xi'an), the reviews did not report relevant city-level data. We therefore conducted city-specific PubMed searches on 12 August 2024 using the following query template (with <CITY> replaced by the specific city name):

```
("Influenza, Human"[Mesh] OR influenza[Title/Abstract] OR ILI[Title/Abstract])
AND (<CITY>[Title/Abstract])
AND (China[Mesh] OR China[Title/Abstract] OR Chinese[Title/Abstract])
AND (seasonality[Title/Abstract] OR seasonal[Title/Abstract]
OR "time series"[Title/Abstract] OR temporal[Title/Abstract])
```

Selection criteria. Found studies were screened against the same eligibility criteria used for RSV: city-level surveillance, ≥ 5 years of continuous observations, and monthly/weekly resolution. Studies were excluded if full text was unavailable, data were incomplete (e.g., interrupted series), or if the surveillance definition was unclear. When multiple eligible series were available for a city (e.g., from the PubMed search), we prioritized series with the longest continuous coverage and specific virological confirmation (e.g., laboratory-confirmed influenza vs. syndromic ILI counts alone).

The final dataset included one influenza time series for each of the eight cities. Data sources and surveillance indicator definitions are provided in Supplementary Table S1.

Table 5: Supplementary Table S1. Surveillance indicator definitions for RSV and influenza (InfV) across included Chinese cities

City	Pathogen	Indicator definition	Clinical setting	Laboratory method	Reference
Beijing	RSV	RSV antigen positivity in ARI inpatients	Inpatient (ARI)	IFA/DFA on NPA	[a]
Beijing	InfV	Influenza-positive samples by subtype	Outpatient (ILI sentinel)	Virus isolation	[b]
Xi'an	RSV	RSV-positive cases / positivity rate	Out-/inpatient (ARI)	Not reported	[c]
Xi'an	InfV	Daily ILI case counts (syndromic)	Outpatient (ILI)	None	[d]
Lanzhou	RSV	RSV positivity; RSV-A genotypes	Inpatient (ARI)	RT-PCR + subgroup PCR + sequencing	[e]
Lanzhou	InfV	Laboratory-confirmed influenza	Out-/inpatient	RT-PCR / isolation / serology	[f]
Wuhan	RSV	RSV positivity in ARTI inpatients	Inpatient (ARTI)	DFA on NP swabs	[g]
Wuhan	InfV	ILI% and influenza-positive specimens	Outpatient (ILI sentinel)	RT-PCR	[h]
Wenzhou	RSV	RSV-positive LRTI inpatients	Inpatient (LRTI)	DFA	[i]
Wenzhou	InfV	Influenza-positive LRTI inpatients	Inpatient (LRTI)	DFA / culture / ELISA	[j]
Suzhou	RSV	RSV positivity in neonatal LRTI inpatients	Inpatient (neonatal LRTI)	DFA on NPA; PCR for atypical agents	[k]
Suzhou	InfV	Influenza positivity in SARI inpatients	Inpatient (SARI)	RT-PCR on NPA	[l]

References

- [1] Diamond, C., Gong, H., Sun, F.Y., Liu, Y., Quilty, B.J., Jit, M., Yang, J., Yu, H., Edmunds, W.J., Baguelin, M., 2022. Regional-based within-year seasonal variations in influenza-related health outcomes across mainland china: a systematic review and spatio-temporal analysis. *BMC Medicine* 20, 58. doi:[10.1186/s12916-022-02269-5](https://doi.org/10.1186/s12916-022-02269-5).
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